# ViT-5: Vision Transformers for The Mid-2020s

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**ONE-LINE TAKEAWAY:** ViT-5 achieves **84.2%** top-1 accuracy on ImageNet-1k classification and **1.84** FID in diffusion generation, outperforming DeiT-III-Base (**83.8%**) and vanilla ViT-based SiT (**2.06**).

## PROBLEM

Vision Transformer architectures have remained largely unchanged since 2021 despite major advancements in language models.

There is a growing architectural gap between vision and language transformers that limits performance and generalizability.

## METHOD

- **Replace LayerNorm** with **RMSNorm** for normalization.
- Use **LayerScale** for activation scaling.
- Apply Rotary Positional Embeddings **(RoPE)** for positional encoding.
- Introduce **register tokens** with **high-frequency RoPE**.
- Avoid **SwiGLU** activations due to over-gating issues.

**FIGURE 1.** Overview of ViT-5 architecture. We conduct in-depth analyses and moder

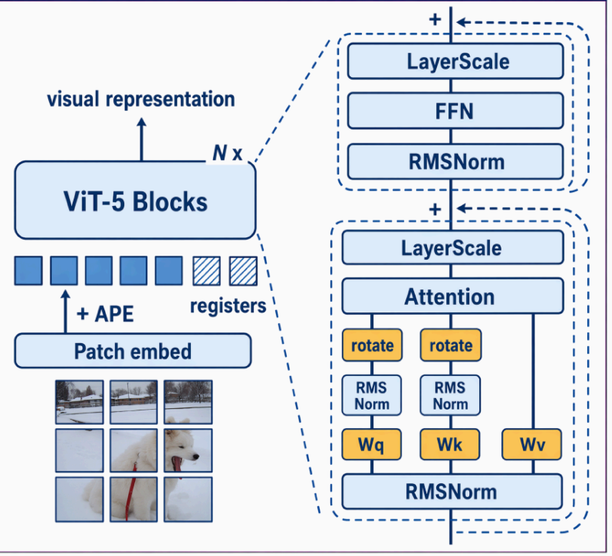
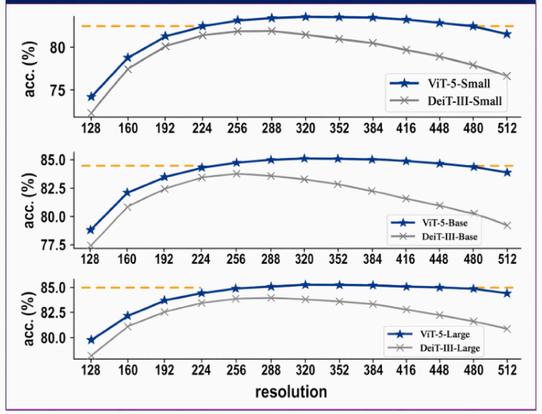



**FIGURE 2.** Undesired invariance when discarding APE. The two images are equivalen

*(a)* original *(b)* patch flipped

**FIGURE 3.** Performance at dynamic resolutions. All models are trained at $224^2$ and




## RESULTS SUMMARY

ViT-5-Base achieves **84.2%** top-1 accuracy on ImageNet-1k classification.

DeiT-III-Base achieves **83.8%** top-1 accuracy on ImageNet-1k classification.

ViT-5 improves SiT diffusion model FID from **2.06** to **1.84.**

ViT-5-Base with post-norm achieves **84.15%** top-1 accuracy (vs. 84.16% with LayerScale).

ViT-5-Base with LayerScale and no SwiGLU achieves **84.16%** top-1 accuracy

ViT-5-Base with registers and high-frequency RoPE achieves **84.86%** top-1 accuracy

## KEY NUMBERS

**84.2%** top-1 accuracy of ViT-5-Base on ImageNet-1k classification.

**83.8%** top-1 accuracy of DeiT-III-Base on ImageNet-1k classification.

**1.84** FID score of SiT diffusion model using ViT-5 backbone.

**2.06** FID score of SiT diffusion model using vanilla ViT backbone.

ViT-5 also serves as a stronger backbone for generative modeling: when plugged into an SiT diffusion framework, it achieves **1.84** FID versus **2.06** with a vanilla ViT backbone.